

NAME: _____ PERIOD: _____ DATE: _____

Two Simulations for Mystery Tokens

AP Statistics · Unit 3 · Topic 3.14 · B: 2026-12-18 · E: 2027-02-01

[STAR] TODAY'S PROBLEM

Suppose a company claims 10,000 mystery tokens are 50% red, 30% teal, and 20% gold. An SRS of 80 has the shown counts.

Leo: “Mix and redraw these same 80 tokens without replacement; recompute χ^2 for 1,000 trials.”

Amara: “For each trial, generate 80 independent colors with probabilities 0.50, 0.30, and 0.20, then recompute χ^2 .” In Amara’s trials, 186 statistics are at least the observed value.

Token sample

Color	Claim	Observed
Red	0.50	32
Teal	0.30	30
Gold	0.20	18

FIRST TAKE — before the video (5 min)

Which simulation can reveal whether the sample’s color discrepancies are unexpected? State what varies in one trial.

[BOOK] BUILD DISCREPANCY FROM THE NULL MODEL (keep on the page)

For model probability p_i and sample size n , category i has expected count np_i . Measure total relative discrepancy with $\chi^2 = \sum (O - E)^2 / E$; for k model categories, $df = k - 1$. A valid null simulation generates a new sample of size n using the claimed probabilities, recomputes χ^2 , and counts simulated values at least as large as observed. Merely rearranging fixed observed labels cannot show sample-to-sample count variation.

Finish after the video:

DOK 1 (a) Calculate the three expected counts under the company's claimed color model.

DOK 2 (b) Compute the χ^2 contributions and total, give df , estimate the upper tail from Amara's trials, and check 10 percent and expected counts.

DOK 3 (c) Choose the simulation that answers the chance-variation question and judge surprise at 5 percent. Compare what each trial fixes or varies, use the simulated tail, explain the rejected output's false assurance, and avoid claiming the model is proved.

Frame: Simulation _____ is valid because one trial allows _____ to vary under _____.

Frame: The tail estimate _____ is _____ 0.05, so _____; the rejected plan would mislead readers by _____.

EXIT — turn this in

One thing the video changed about my first take: _____